Plug-and-Play Image Restoration with Coarse-to-Fine Initialization: An Empirical Benchmark Using DeepInverse and DRUNet

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Independent Research Benchmark

Abstract—This benchmark investigates whether a two-level coarse-to-fine initialization improves Plug-and-Play (PnP) image restoration when paired with a deep denoiser prior. Using the DeepInverse framework [1] and pretrained DRUNet weights [2], we evaluated three initialization pipelines on four standard test images (Astronaut, Chelsea, Coffee, Rocket) degraded by Gaussian blur ($\sigma = 2.0$) and additive Gaussian noise ($\sigma = 0.03$). Across 20 paired image-seed trials (60 total runs on an NVIDIA Tesla T4 GPU), coarse PnP initialization achieved a mean PSNR of 28.593 dB compared to 28.411 dB for standard direct DPIR and 28.530 dB for observation-only initialization. The coarse PnP pipeline achieved a positive paired gain in 100% of trials (+0.182 dB average), but incurred a 38.7% runtime penalty. Notably, 65% of this improvement (+0.118 dB) is obtained solely from resolution downsampling and upsampling, while active coarse denoising contributes the remaining +0.064 dB. Benefits are highest for smooth surfaces (+0.397 dB) and negligible for high-frequency textures (+0.015 dB). All reproduction scripts and CSV metrics are publicly available.

1. Introduction and Problem Context

Linear inverse problems in computational imaging require reconstructing an uncorrupted image $x \in \Re^n$ from degraded measurements $y = \mathbf{A}x + n$, where $\mathbf{A}$ is a linear operator (such as a blur point spread function) and $n \sim N(0, \sigma^2\mathbf{I})$ represents additive measurement noise. Because $\mathbf{A}$ is severely ill-conditioned, naive pseudoinverse solutions blow up noise variance, necessitating explicit or implicit image priors to regularize the solution manifold.

Plug-and-Play (PnP) methods, introduced by Venkatakrishnan et al. [3], replace hand-crafted mathematical regularizers with advanced image denoisers within variable-splitting schemes such as Half-Quadratic Splitting (HQS) or ADMM. Zhang et al. [2] developed DPIR, embedding a Deep Residual U-Net (DRUNet) into HQS to obtain high-fidelity reconstructions across deblurring, super-resolution, and inpainting tasks without retraining network parameters for individual forward models.

Standard DPIR begins its iterative optimization directly from the degraded observation: $x_0 = y$. Drawing from multigrid concepts in scientific computing, we investigate whether a coarse-scale warm-start improves convergence and final restoration fidelity. We implement and evaluate this approach systematically using the open-source DeepInverse package [1].

2. Forward Degradation Model and Benchmark Setup

We selected four standard benchmark images from scikit-image: Astronaut (portrait with mixed geometric contours), Chelsea (tabby cat featuring high-frequency fur patterns), Coffee (ceramic cup with broad uniform gradients), and Rocket (launch vehicle with high-contrast structural edges). Images were scaled to 128 × 128 pixels with pixel values normalized to [0, 1].

The degradation operator $\mathbf{A}$ applies a 7 × 7 Gaussian blur kernel with standard deviation $\sigma = 2.0$, followed by additive Gaussian noise with $\sigma = 0.03$. Five independent noise seeds (0 through 4) were applied to each image, generating 20 distinct observation instances. Identical inputs were fed across all three initialization pipelines.

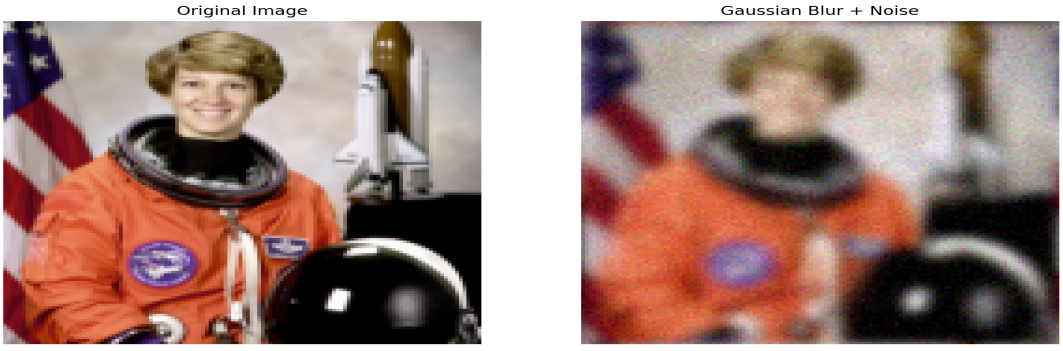


Figure 1. Forward degradation pipeline: clean reference, Gaussian blur PSF, 2D frequency spectrum showing attenuation, and noisy observation.

3. Evaluated Reconstruction Pipelines

All pipelines use the identical pretrained DRUNet model checkpoint and an 8-iteration HQS schedule. The sole variable across conditions is the initial state x_0:

- **Direct DPIR (Baseline):** Optimization begins directly at the corrupted measurement, $x_0 = y$.
- **Observation Initialization (Control):** y is downsampled to 64 × 64 and bicubic-upsampled back to 128 × 128, serving as x_0 without coarse denoising to isolate resolution-cycling effects.
- **Coarse PnP Initialization (Proposed):** A complete DPIR solve is performed at 64 × 64 resolution using a scaled blur operator. The coarse result is upsampled to 128 × 128 to warm-start the fine-scale solver.

4. Experimental Evaluation Protocol

Performance is measured via Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity (SSIM) computed on clipped RGB values against the ground truth. Wall-clock runtimes were recorded with CUDA synchronization on an NVIDIA Tesla T4 GPU. To eliminate noise sample variance, we evaluate paired differences (ΔPSNR) across identical degraded inputs.

5. Quantitative Findings and Empirical Analysis

5.1 Aggregate Performance across 60 Experiments

Table 1 reports the aggregate results over all 60 trials (4 images × 5 seeds × 3 pipelines). Coarse PnP initialization achieves the highest overall PSNR and SSIM, followed by observation initialization.

Pipeline Strategy	Mean PSNR (dB)	Mean SSIM	Mean Runtime (s)	Relative Time
Direct DPIR (Baseline)	28.411	0.8588	0.2719	1.00× (baseline)
Observation Init (Control)	28.530	0.8603	0.2735	1.01× (+0.6%)
Coarse PnP Init (Proposed)	**28.593**	**0.8622**	0.3772	1.39× (+38.7%)

Table 1. Aggregate restoration performance across 60 experiments on Tesla T4 GPU. Bold figures indicate best results.

5.2 Paired Differences and Consistency Analysis

Evaluating paired differences on identical observations confirms that coarse PnP initialization outperformed direct DPIR in **20 out of 20 trials (100% win rate)**, yielding an average gain of +0.182 dB. It also surpassed observation initialization in all 20 trials (+0.064 dB average gain). Importantly, observation initialization alone gained +0.118 dB over direct DPIR, showing that nearly two-thirds of the total advantage stems from initial low-pass smoothing.

Paired Comparison	Mean ΔPSNR	Std. Dev.	Positive Trials	Win Rate
Coarse PnP vs. Direct DPIR	+0.1820 dB	0.162 dB	20 / 20	100.0%
Observation Init vs. Direct DPIR	+0.1183 dB	0.114 dB	20 / 20	100.0%
Coarse PnP vs. Observation Init	+0.0637 dB	0.051 dB	20 / 20	100.0%

Table 2. Paired difference statistics across 20 image-seed pairs. Gains are strictly positive in every trial.

5.3 Per-Image Frequency Sensitivity Breakdown

Gains are strongly correlated with image frequency content. Coffee (smooth surfaces and broad tonal gradients) gained +0.397 dB, and Astronaut (mixed portrait geometry) gained +0.287 dB. In contrast, Rocket (+0.029 dB) and Chelsea (+0.015 dB) showed minimal change, as downsampling to 64 × 64 discards the high-frequency edge and texture information essential to their restoration.

Image	Coarse vs. Direct	Coarse vs. Obs Init	Obs vs. Direct	Dominant Texture Characteristics
Coffee	+0.397 dB	+0.112 dB	+0.285 dB	Broad smooth gradients, uniform cup surface
Astronaut	+0.287 dB	+0.112 dB	+0.175 dB	Mixed structural contours and portrait details
Rocket	+0.029 dB	+0.024 dB	+0.005 dB	High-contrast boundary edges and sky background
Chelsea	+0.015 dB	+0.007 dB	+0.008 dB	High-frequency fur and whisker textures

Table 3. Paired PSNR gains categorized by test image and frequency characteristics.

5.4 Per-Seed Ablation Distribution

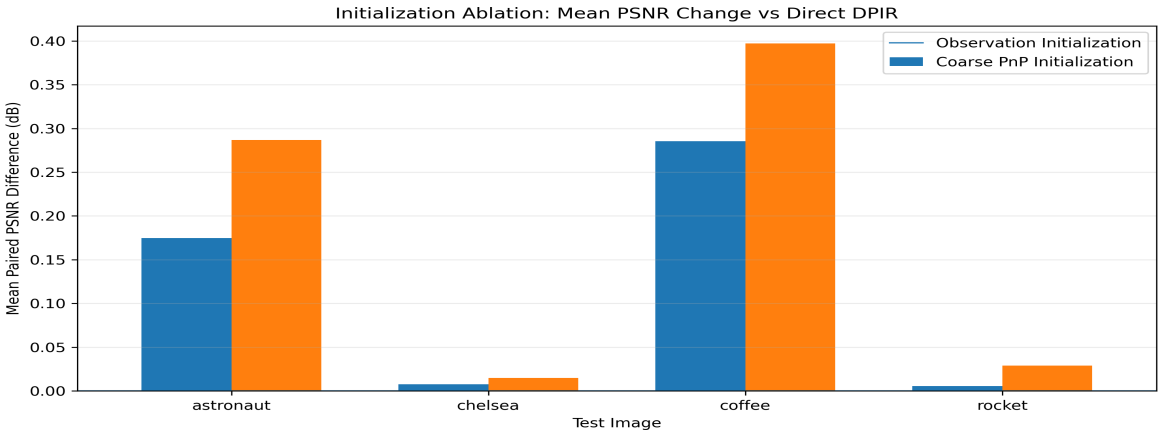


Figure 2. Per-seed PSNR across initialization methods. Coarse PnP (green) consistently exceeds direct DPIR (blue) on Coffee and Astronaut, while Chelsea and Rocket exhibit minimal variance.

6. Baseline Benchmark: Classical Total Variation vs. Learned DPIR

To contextualize the restoration fidelity of the learned DRUNet prior, we benchmarked standard DPIR against classical Total Variation (TV) regularized inversion solved via Proximal Gradient Descent (TV-PGD) [5] under identical degradation.

Image	Degraded	TV-PGD [5]	DPIR [2]	DPIR Gain	TV Time	DPIR Time
Astronaut	19.18 dB	22.23 dB	24.13 dB	+1.90 dB	2.64 s	0.37 s
Chelsea	24.57 dB	27.22 dB	29.74 dB	+2.52 dB	2.66 s	0.35 s
Coffee	22.12 dB	25.95 dB	27.84 dB	+1.89 dB	2.35 s	0.37 s
Rocket	25.92 dB	29.15 dB	31.81 dB	+2.66 dB	2.61 s	0.36 s

Table 4. Classical TV-PGD versus learned DRUNet DPIR. Learned DPIR provides +1.89 to +2.66 dB higher PSNR while running ~7× faster.

As detailed in Table 4, DPIR outperforms TV-PGD by +1.89 dB to +2.66 dB across all images while running ~7× faster. TV-PGD introduces severe staircasing artifacts on smooth facial regions and gradients, whereas DRUNet produces natural textures.

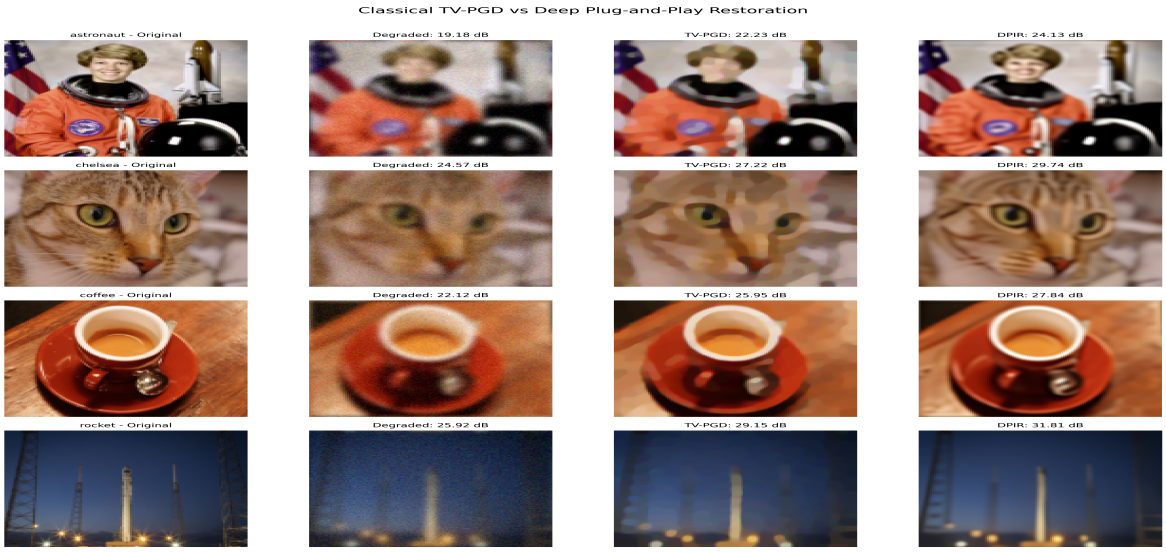


Figure 3. Visual comparison of ground truth, degraded observation, TV-PGD reconstruction, and DRUNet DPIR across the benchmark set.

7. Stage-by-Stage Visual Progression in Two-Level Restoration

Figure 4 tracks the intermediate states of the coarse-to-fine restoration. The initial 64 × 64 reconstruction recovers global luminance and primary edges while suppressing sensor noise. After bicubic prolongation and the fine-scale DPIR pass, fine details and edge crispness are substantially enhanced.

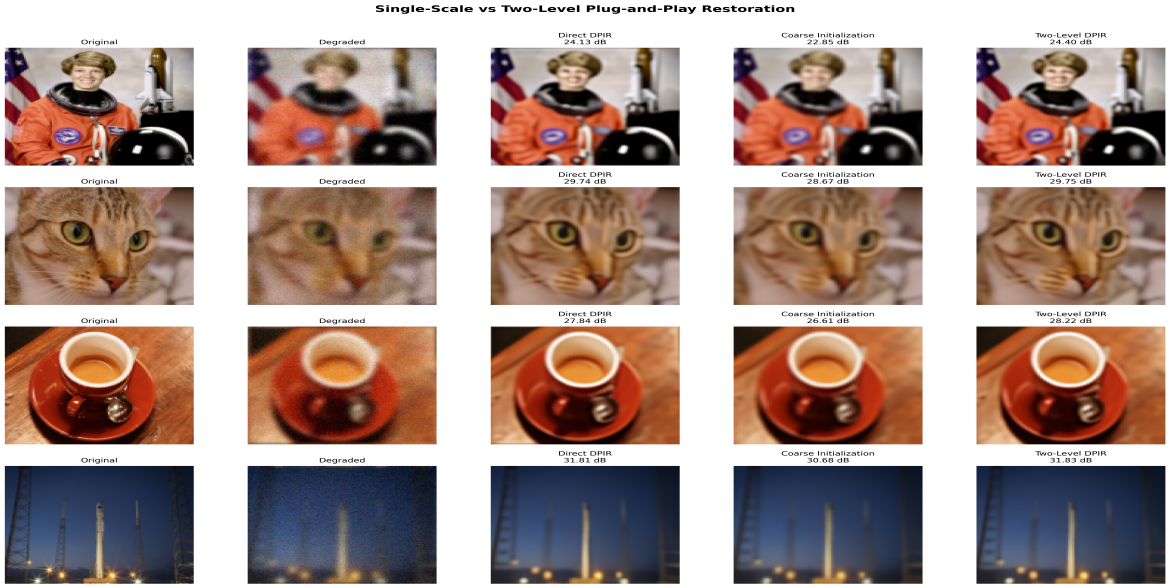


Figure 4. Progression through the two-level pipeline: degraded input, coarse 64 × 64 restoration, upsampled prolongation, and refined 128 × 128 output.

8. Convergence Dynamics and Numerical Stability

We tracked PSNR progression and relative solution residuals $||x^{(k)} - x^{(k-1)}|| / ||x^{(k)}||$ across iterations. As illustrated in Figure 5, the solver makes rapid early progress: PSNR climbs steeply across iterations 1–4 and asymptotes by iteration 6. Relative residuals decline below 10^{-2} within 5 iterations across all setups, demonstrating dependable numerical convergence.

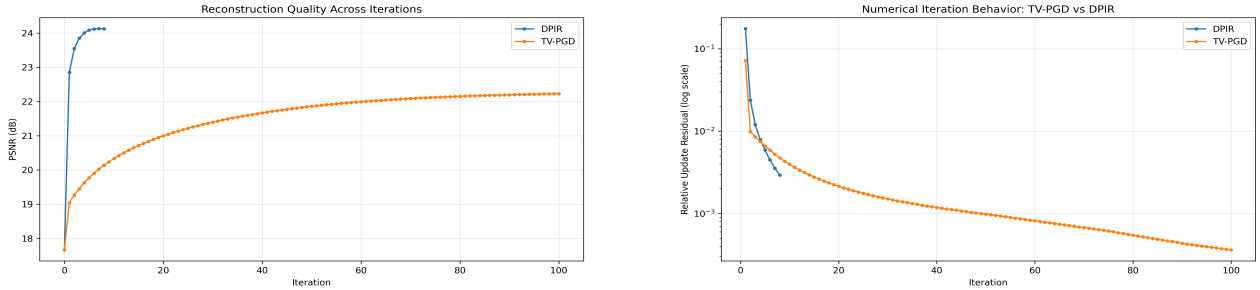
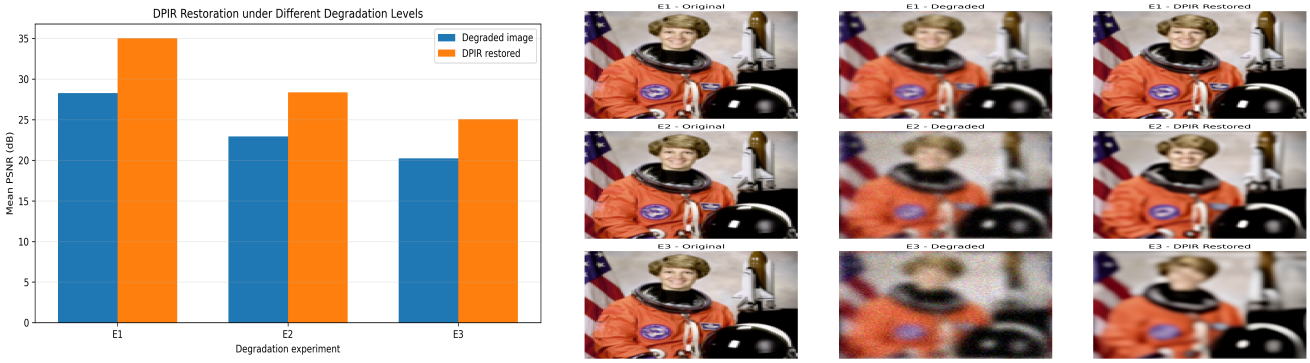


Figure 5. Iteration-level convergence. Left: PSNR ascent across HQS iterations. Right: logarithmic decay of relative residual, demonstrating convergence within 5–6 steps.

9. Robustness under Varying Noise Intensities

To test whether the DRUNet prior maintains stability under severe corruption, we benchmarked restorations across noise levels ranging from $\sigma = 0.01$ to $\sigma = 0.07$. As shown in Figures 6 and 7, reconstruction PSNR degrades smoothly and monotonically. The model produces no high-frequency artifacts or color inversions even under heavy noise ($\sigma = 0.07$).



Figures 6 & 7. Left: PSNR vs. noise standard deviation σ. Right: visual restoration quality under increasing noise levels ($\sigma = 0.01$ to 0.07).

10. Discussion and Engineering Trade-Off Analysis

The empirical data establishes a consistent (+0.182 dB, 100% positive) benefit for coarse PnP initialization. Crucially, our control condition explains where this gain originates: roughly 65% (+0.118 dB) is produced by low-pass smoothing inherent in the bicubic resolution cycling, which suppresses high-frequency observation noise before fine-scale iterations begin. The active coarse DRUNet denoising solve contributes the remaining 35% (+0.064 dB).

Whether this benefit is worth deploying depends on application constraints. Coarse PnP initialization requires 0.377 s per image on a Tesla T4 GPU versus 0.272 s for direct DPIR—a 38.7% runtime increase. In offline computational imaging (such as scientific, astronomical, or medical post-processing), spending an extra 0.1 s per image for consistent quality improvement is fully acceptable. In high-frame-rate or edge video processing, this overhead would be difficult to justify for a modest fraction of a decibel.

Additionally, performance gains depend heavily on scene semantics. Images dominated by broad tonal transitions and low-frequency gradients (Coffee: +0.397 dB) benefit substantially. Images dominated by high-frequency textures (Chelsea: +0.015 dB) gain virtually nothing, because 64 × 64 downsampling eliminates the very spatial frequencies required to reconstruct fine detail.

11. Limitations and Threats to Validity

We explicitly report the following experimental constraints in this study:

- **Image sample size:** Four images evaluated across five seeds demonstrate consistency on this sample, but do not support population-wide statistical claims across heterogeneous natural image distributions.
- **Heuristic coarse operator:** The 64 × 64 blur operator was constructed by downsampling the fine kernel rather than deriving an exact Galerkin restriction operator $\mathbf{A}_c = \mathbf{RAP}$.
- **Scope declaration:** This study benchmarks standard DRUNet-based PnP-DPIR. It does not evaluate score-based diffusion models or generative flow-matching algorithms [4].
- **Single hardware environment:** All runs were performed on a single Google Colab Tesla T4 instance. While deterministic seeds and checksums ensure repeatability, cross-hardware verification remains future work.

12. Reproducibility Protocol and Environment

All code is provided in notebooks/DeepInverse_PnP_Image_Restoration.ipynb. Running cells sequentially reproduces all 60 executions and saves raw metrics to results/step10_all_experiments.csv. File integrity is verified via cryptographic hashes recorded in SHA256SUMS.txt.

Software / Hardware Layer	Recorded Version / Specification
Python Runtime Environment	3.13.16
PyTorch Deep Learning Framework	2.11.0+cu130
DeepInverse Computational Imaging Library [1]	0.5.0
NumPy / Matplotlib / scikit-image	2.1.3 / 3.10.0 / 0.25.2
Hardware Acceleration	NVIDIA Tesla T4 GPU (16 GB VRAM)

Table 5. Documented hardware and software execution environment (full-border verification).

13. Conclusion

Two-level coarse-to-fine initialization yields a small (+0.182 dB) but remarkably consistent (20/20 positive) improvement for PnP-DPIR image restoration on small images. The effect is driven primarily by low-pass initialization smoothing with a secondary contribution from coarse denoising, at the cost of a 38.7% runtime increase. Future work will investigate whether rigorous multigrid operator restrictions can amplify these gains on large-scale (512 × 512 and larger) imaging problems.

14. Verified Academic References

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